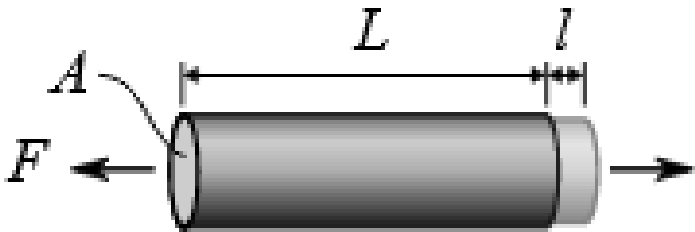


STM - Exercise 1

Exercise 1

Use figure 1 to:

- a) define which test and which class of materials is referred to and why;
- b) calculate elastic modulus,
- c) Calculate the yield strength (0.2 % offset) and the tensile strength,
- d) Calculate the elongation at fracture for a 50 mm long specimen (write on curves data you use).
- e) Calculate the length of the specimen when you keep it loaded in A.



$$\sigma = \frac{F}{A}$$

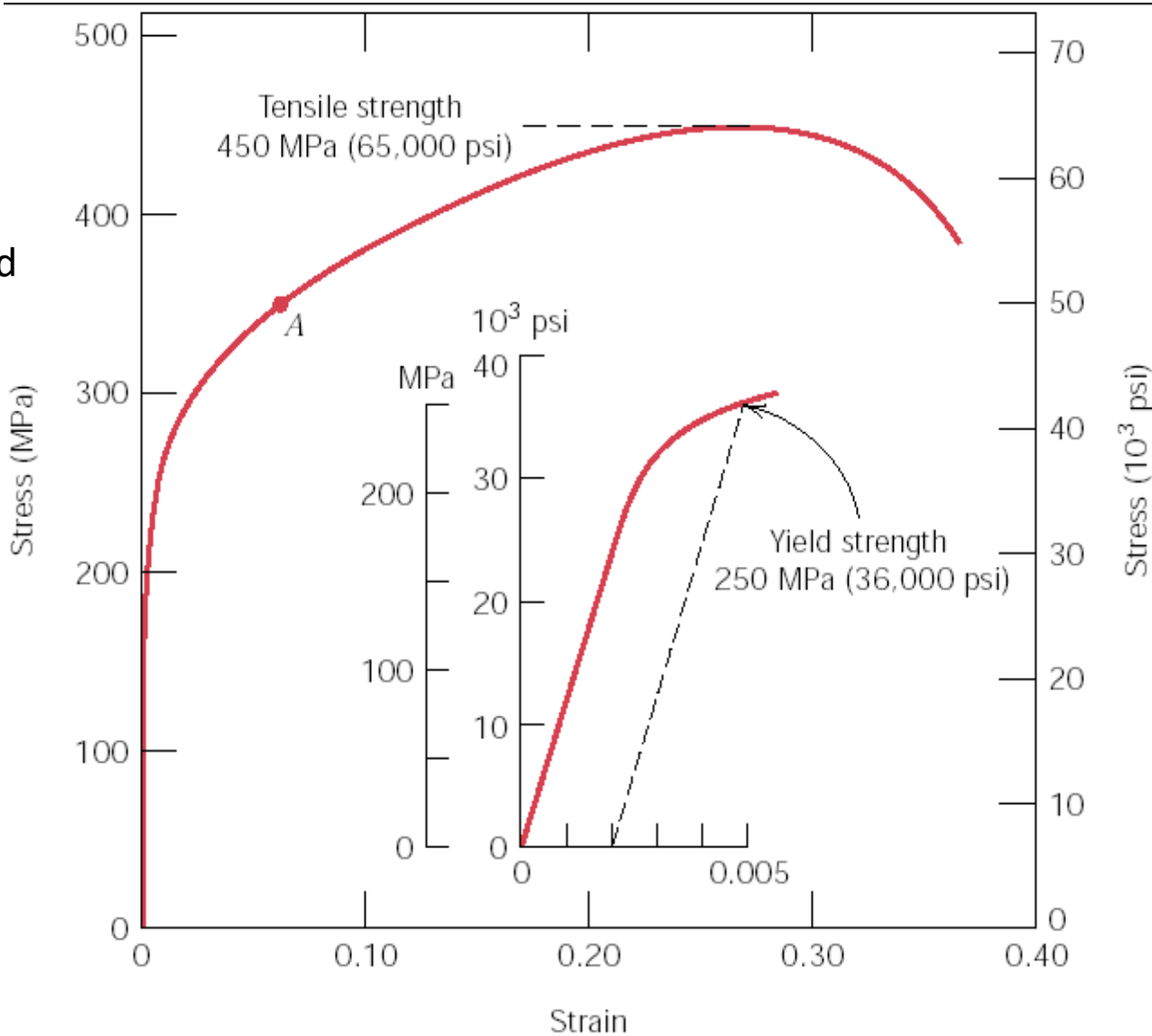
where

- σ stress [Pa]
- F applied force [N]
- A cross-sectional area [m²]

$$\varepsilon = \frac{\Delta L}{L_0}$$

where

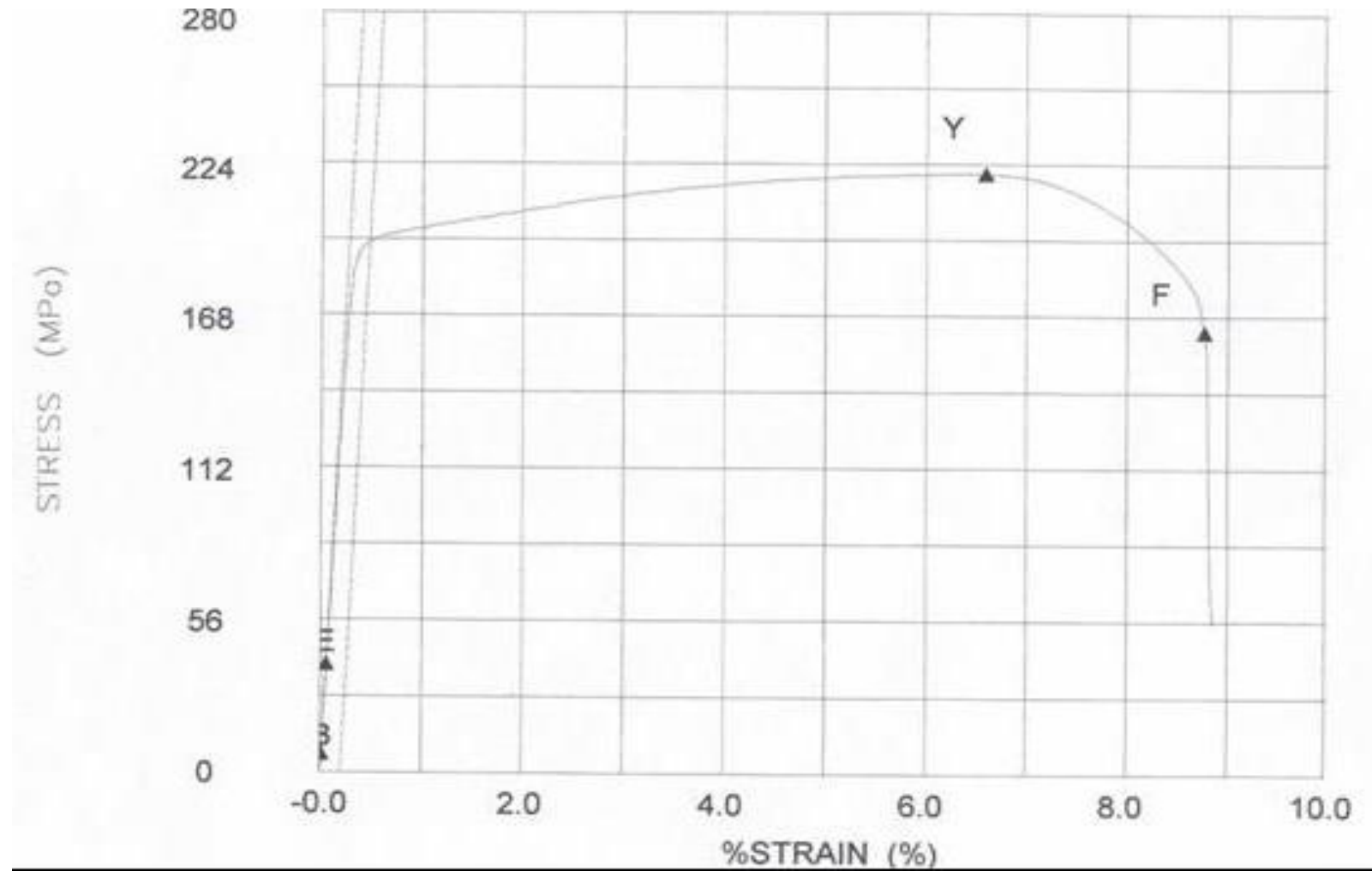
- ε strain
- ΔL total elongation [m]
- L_0 original length [m]



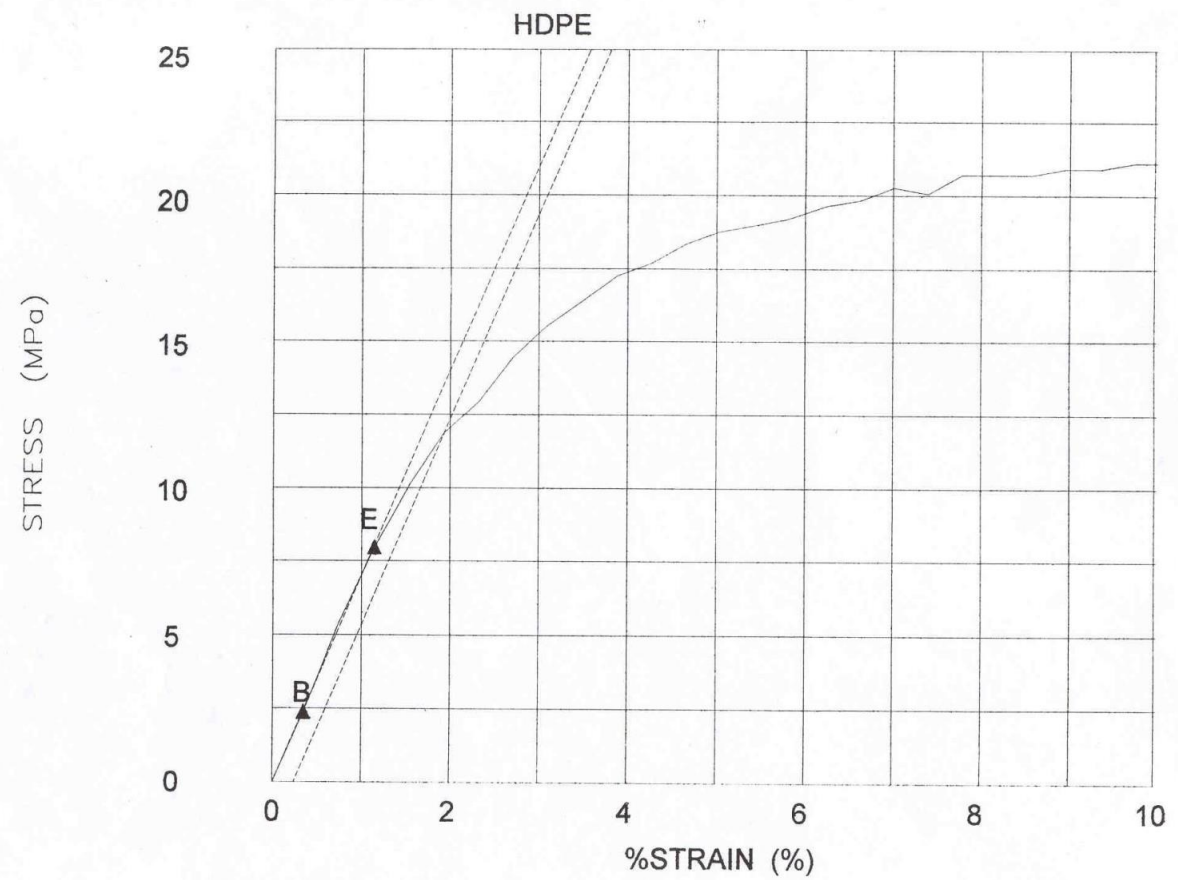
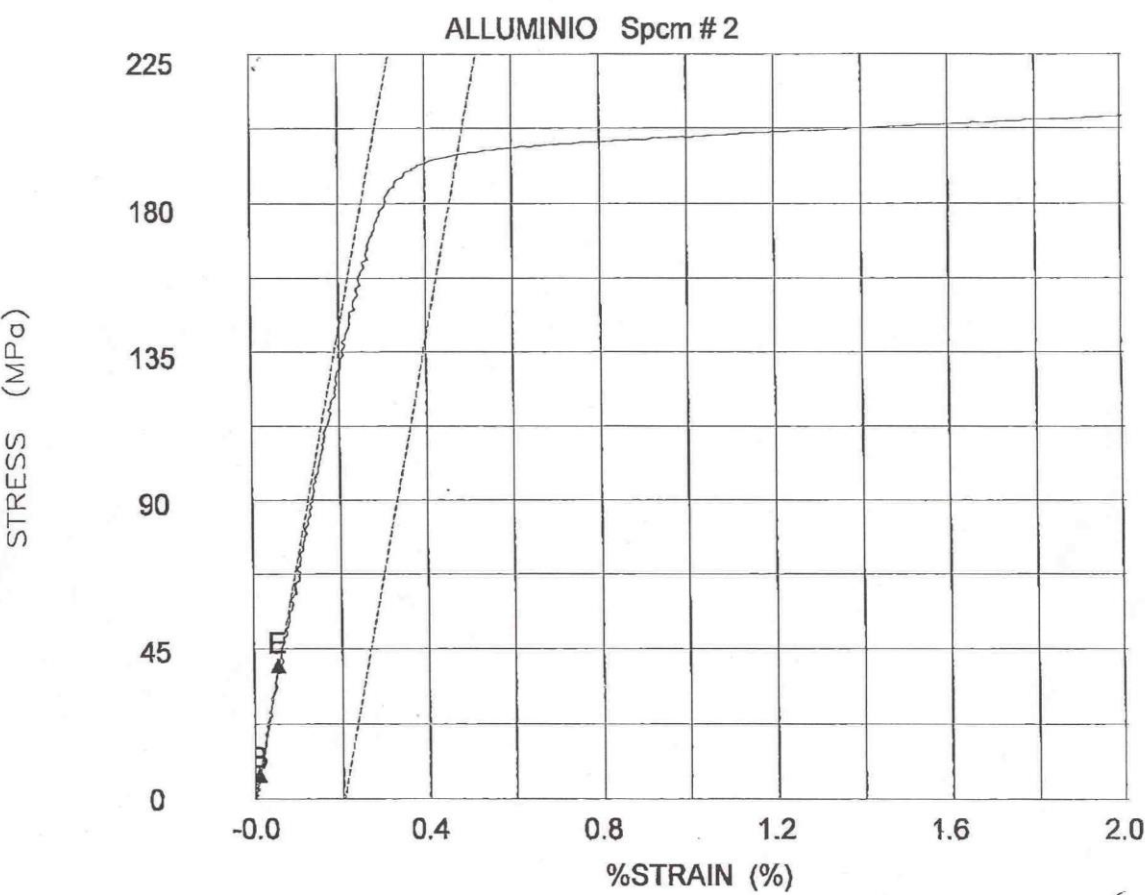
Exercise 2: Use the graph below to define

- which test and which class of materials is referred to and why;
- to calculate elastic modulus,
- To calculate the yield strength (0,2 % strain) and the tensile strength,
- To calculate the elongation at fracture for a 50 mm long specimen;
- To calculate the elongation of this sample when loaded at 200 MPa, then unloaded.

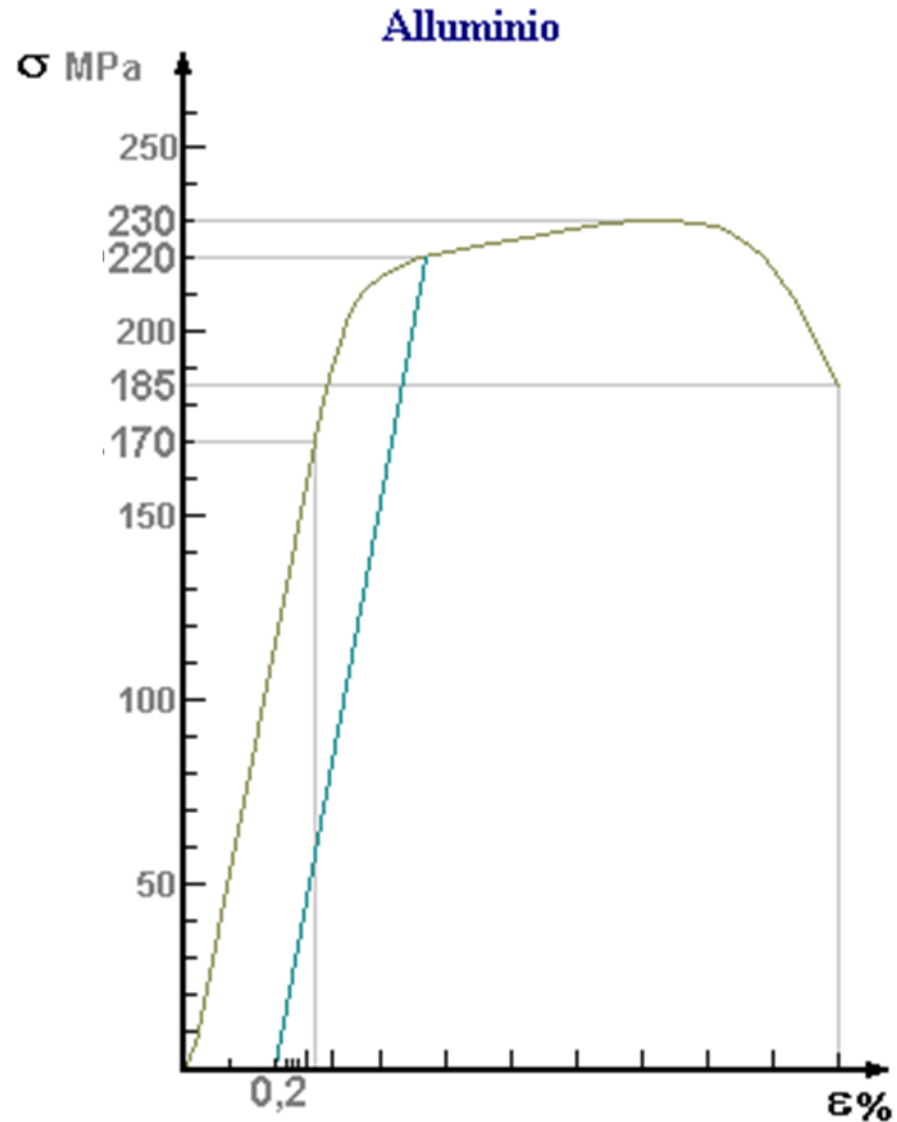
(write on graph stress and strain
you use to calculate).



Exercise 3: Determine the Elastic Modulus of Aluminum and High Density Polyethylene (HDPE) from the graphs below. Comment the obtained results.



Quiz 1

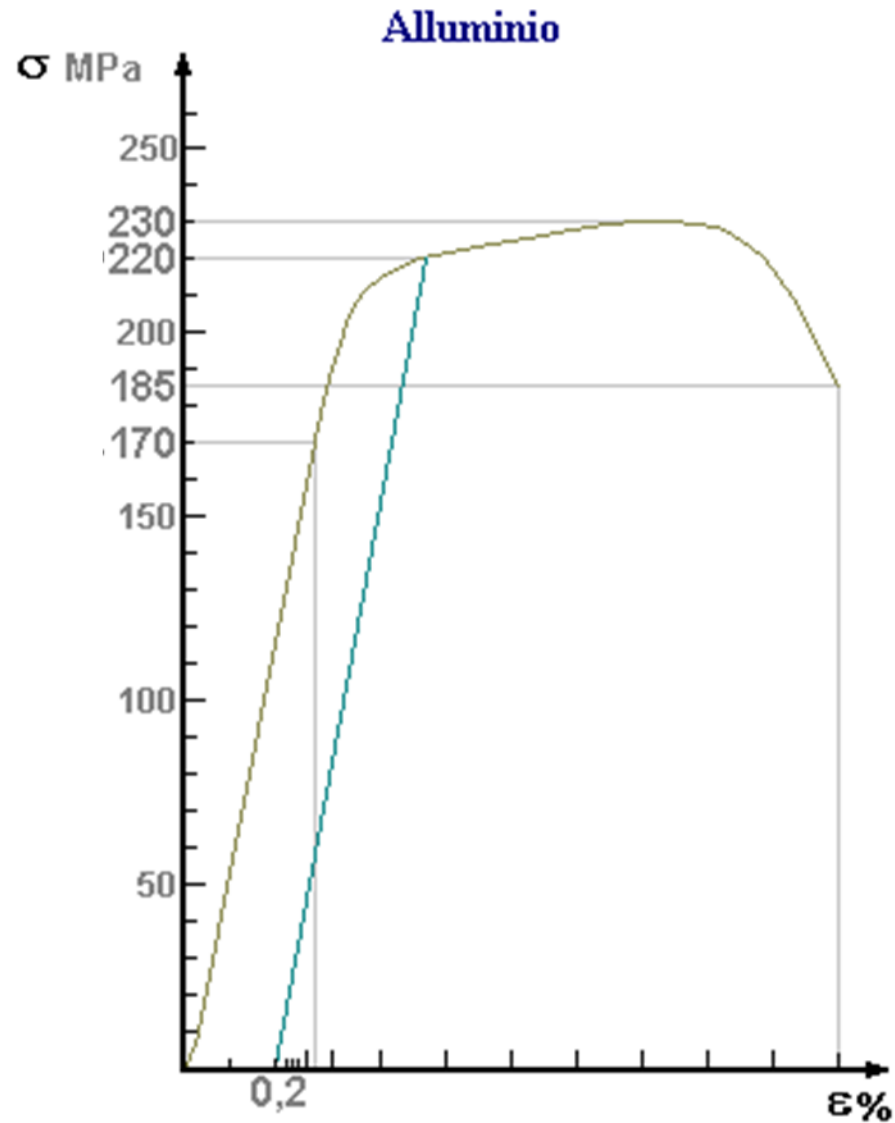


Consider the σ - ϵ curve reported in the figure.

The yield strength (0.2 % offset) is:

- a) 230 Mpa
- b) 220 Mpa
- c) 185 MPa

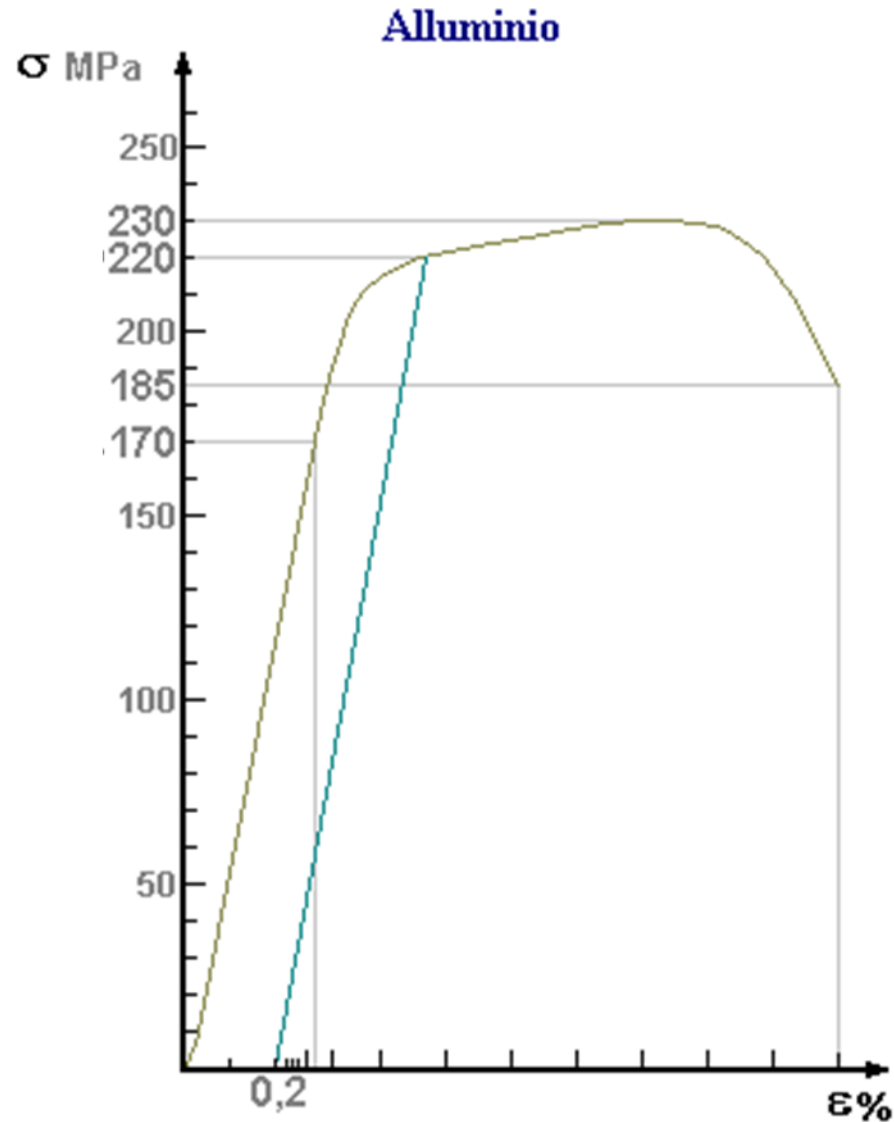
Quiz 2



The σ - ϵ curve in figure comes from the tensile test of an aluminum specimen. Select the right sentence (only one):

- a) The specimen underwent plastic deformation
- b) The specimen underwent fragile fracture
- c) The specimen did not have permanent deformations

Quiz 3



The σ - ϵ curve in figure comes from the tensile test of an aluminum specimen. Select the right sentence (only one):

- a) The specimen had only elastic deformation
- a) The specimen underwent necking
- a) The yield strenght was not reached in the test